Lesson 3: V=IR

Electricity and Magnetism | GCSE Combined Science

Gemini Notebook

Starter: Retrieval Grid (5 Minutes)

State Ohm's law equation linking V, I, and R.
[Voltage, current, resistance]

V = I × R (or R = V / I)

What is the unit of electrical resistance?
[Greek letter omega]

Ohm (Ω)

If resistance increases at constant voltage, what happens to current?
[Inversely proportional]

Current decreases

How does doubling the length of a wire affect its resistance?
[Twice as many collisions]

Resistance doubles (directly proportional)

Why should the power pack be switched off between readings?
[Temperature control]

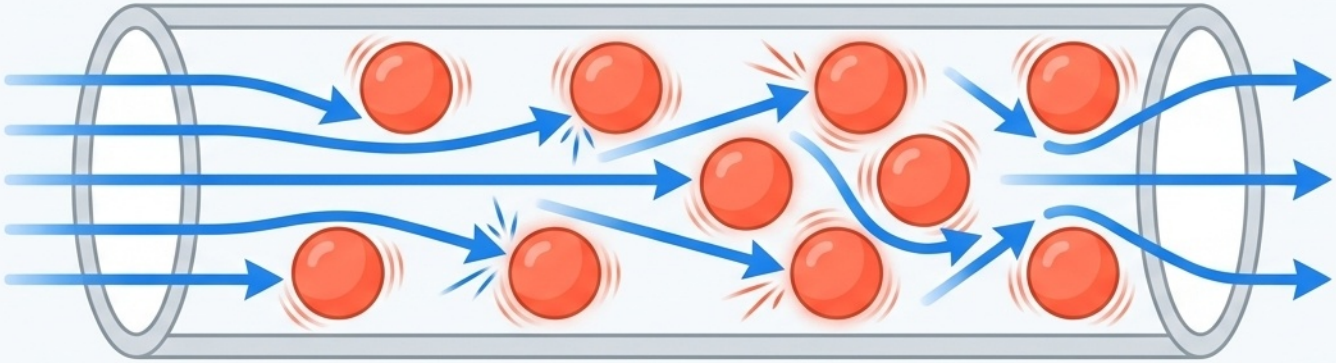
To prevent the wire heating up (which changes resistance)

Calculate V if I = 2 A and R = 5 Ω.
[V = I × R]

10 V (V = 2 × 5)

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Defining Resistance



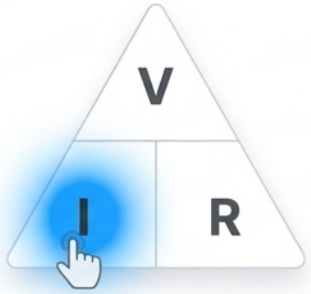
- **Definition:** Opposition to the flow of electrical current.
- **The Cause:** Electrons physically colliding with vibrating metal ions.
- **The Result:** Energy is transferred, often as heat.

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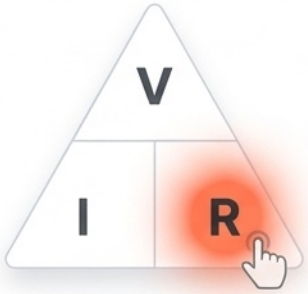
The Rule: Ohm's Law



$V = I \times R$



$I = V / R$



$R = V / I$

- **V** = Potential Difference (Volts, V)
- **I** = Current (Amperes, A)
- **R** = Resistance (Ohms, Ω)

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Calculation Race (Mini-Whiteboards Ready!)

Find V

Current = **3 A**
Resistance = **4 Ω**

[Reveal Answer: 12 V]

Find R

Potential Difference = **10 V**
Current = **2 A**

[Reveal Answer: 5 Ω]

Find I

Potential Difference = **24 V**
Resistance = **8 Ω**

[Reveal Answer: 3 A]

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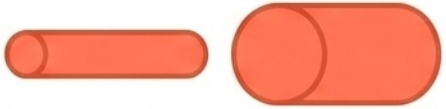
Factors Affecting Resistance

Length



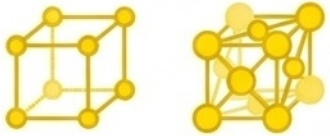
Longer wire = greater resistance.

Area



Thicker wire = lower resistance.

Material



Different atomic structures alter flow.

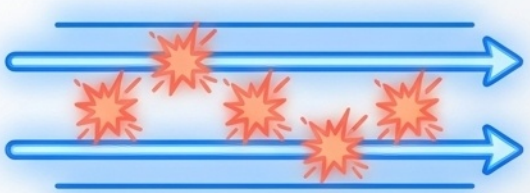
Temperature



Hotter wire = greater resistance.

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Deep Dive: Length vs. Resistance



Wire A (Length L)



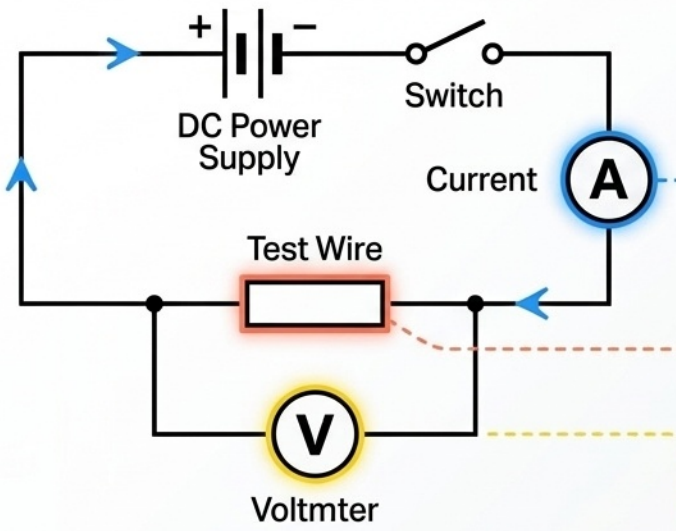
Wire B (Length 2L)

- **The Principle:** Double the length = double the collisions.
- **The Physics Term:** Resistance is directly proportional to length.
- **The Lab Goal:** We will prove this today using nichrome wire.

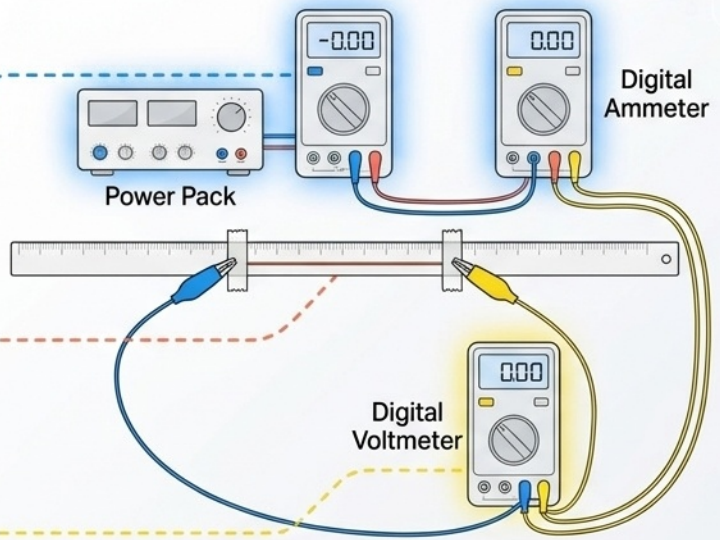
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Practical Setup: Resistance of a Wire

Circuit Diagram

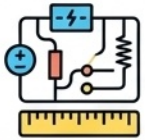


Physical Desk Setup



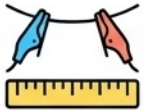
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Mandatory Practical: The Workflow



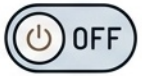
Setup

Attach nichrome wire to metre ruler. Build circuit.



Measure

Attach crocodile clips at 20 cm. Record V and I.



Rest

Switch off power pack. Calculate R = V / I.



Repeat

Move clips to 40, 60, 80, and 100 cm. Repeat steps.

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Quality Control: Why Switch Off?

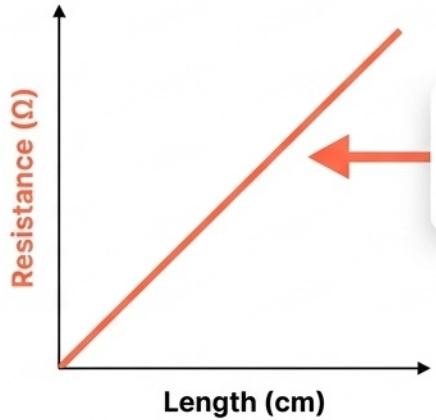


- **The Problem:** Current causes the wire to heat up.
- **The Impact:** Heat increases resistance (adding a second variable).
- **The Solution:** Switch off between readings to keep temperature constant.

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Data Synthesis: Plotting Your Results

Length (cm)	Potential Difference (V)	Current (A)	Resistance (Ω)



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Pair & Share Checkpoint

3 Minutes

Conceptual

Using the collision model, explain why a thicker wire has lower resistance than a thin wire.

Practical Evaluation

If a student forgets to switch off the power pack between readings, how will their final graph look different?

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Lesson Summary

1. **V = I × R** (Ohm's Law connects Potential Difference, Current, and Resistance).
2. Resistance is the physical opposition to current flow (collisions).
3. Length and resistance are directly proportional (Double length = double resistance).

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